

“Año de la esperanza y el fortalecimiento de la democracia”

UNIVERSIDAD PRIVADA SAN JUAN BAUTISTA

FACULTAD DE CIENCIAS DE LA SALUD



ACTIVIDAD 13

**ANALISIS DE COMPONENTES PRINCIPALES (PCA) Y ANALISIS
DE CONGLOMERADOS (K-MEANS)**

DOCENTE:

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CURSO:

Sistematización y métodos estadísticos

ALUMNA:

Muñoz Arevalo Milagros Lucero

2026

IMPORTAR LA BASE (PACIENTES)

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Render on Save Render

Source Visual Normal Format Insert Table Outline

Package markdown required but is not installed. Install Don't Show Again

data.frame

```
'data.frame': 30 obs. of 9 variables:
 $ Paciente      : chr "p1" "p2" "p3" "p4" ...
 $ Edad          : int 24 27 30 32 35 38 42 45 48 50 ...
 $ IMC           : num 21.5 22.1 23 24 24.5 25.2 26 27 27.5 28 ...
 $ Glucosa       : int 88 92 95 98 102 108 115 120 128 135 ...
 $ Presion_Sistolica : int 110 112 115 118 120 122 126 128 130 132 ...
 $ Presion_Diastolica : int 70 72 75 76 78 80 82 84 85 86 ...
 $ Colesterol    : int 165 170 175 180 185 190 195 200 205 210 ...
 $ Saturacion    : int 99 98 98 97 98 97 97 96 96 96 ...
 $ Frecuencia_Cardiaca : int 68 70 72 74 75 76 78 80 82 84 ...
 Paciente      :      Edad
Length:30      Min.      :24.00
```

Quarto Running Code

Environment History Connections Tutorial

Import Dataset 220 MiB

R Global Environment

Data

datos	30 obs. of 9 variables
-------	------------------------

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

C:\Users\LENOVO\OneDrive\Escritorio\LUCERO SISTEMATIZACION - SABADO TARDE

Name	Size
.Rhistory	27 B
DATA	
LUCERO SISTEMATIZACION.Rproj	218 B
pacientes_caidas.csv	614 B
pacientes.csv	1.1 K
QUARTO	
SCRIPT	

Console Terminal Background Jobs

R 4.5.3 C:\Users\LENOVO\OneDrive\Escritorio\LUCERO SISTEMATIZACION - SABADO TARDE

```
> # Importar la base de datos
> datos <- read.csv("pacientes.csv", header = TRUE)
> # Visualizar las primeras filas
> head(datos)
> # Ver estructura
> str(datos)
'data.frame': 30 obs. of 9 variables:
 $ Paciente      : chr "p1" "p2" "p3" "p4" ...
 $ Edad          : int 24 27 30 32 35 38 42 45 48 50 ...
 $ IMC           : num 21.5 22.1 23 24 24.5 25.2 26 27 27.5 28 ...
 $ Glucosa       : int 88 92 95 98 102 108 115 120 128 135 ...
 $ Presion_Sistolica : int 110 112 115 118 120 122 126 128 130 132 ...
 $ Presion_Diastolica : int 70 72 75 76 78 80 82 84 85 86 ...
 $ Colesterol    : int 165 170 175 180 185 190 195 200 205 210 ...
 $ Saturacion    : int 99 98 98 97 98 97 97 96 96 96 ...
 $ Frecuencia_Cardiaca : int 68 70 72 74 75 76 78 80 82 84 ...
```

PARA REALIZAR EL PCA

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Render on Save Render

Source Visual Normal Format Insert Table Outline

Package markdown required but is not installed. Install Don't Show Again

pca <- prcomp(variables)

Resultados

summary(pca)

Cargas de los componentes

pca\$rotation

Coordenadas de los pacientes

pca\$x

Gráfico

biplot(pca)

Quarto Running Code

Environment History Connections Tutorial

Import Dataset 177 MiB

R Global Environment

Data

datos	30 obs. of 9 variables
pca	List of 5
variables	num [1:30, 1:8] -2.05 -1.86 -1.67 -1.54 -1.35...

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

C:\Users\LENOVO\OneDrive\Escritorio\LUCERO SISTEMATIZACION - SABADO TARDE

Name	Size	Modified
.Rhistory	27 B	May 1, 2026, 1:05 AM
DATA		
LUCERO SISTEMATIZACION.Rproj	218 B	Jun 28, 2026, 3:06 AM
pacientes_caidas.csv	614 B	Jun 27, 2026, 3:02 PM
pacientes.csv	1.1 KB	Jun 27, 2026, 2:40 PM
QUARTO		
SCRIPT		

Console Terminal Background Jobs

R 4.5.3 C:\Users\LENOVO\OneDrive\Escritorio\LUCERO SISTEMATIZACION - SABADO TARDE

```
[17,] 0.032768338
[18,] 0.005736718
[19,] -0.008266589
[20,] -0.009248651
[21,] 0.005485933
[22,] 0.014832752
[23,] 0.008462924
[24,] 0.004026772
[25,] 0.013373590
[26,] 0.007075831
[27,] -0.011071223
[28,] -0.025764187
[29,] -0.024740505
[30,] 0.018758672
> # Gráfico
> biplot(pca)
```

PASO 1 (PACIENTES CAIDAS)

LUCERO SISTEMATIZACION - SABADO TARDE - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Render on Save ABC Render

Source Visual B T Normal Format Insert Table Outline

Package markdown required but is not installed. Install Don't Show Again

```
datos <- read.csv("pacientes_caidas.csv")
head(datos)
summary(datos)
```

Quarto Running Code

Environment History Connections Tutorial

R Global Environment

Data

datos	20 obs. of 8 variables
pca	List of 5
variables	num [1:30, 1:8] -2.05 -1.86 -1.67 -1.54 -1.35...

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

C > Users > LENOVO > OneDrive > Escritorio > LUCERO SISTEMATIZACION - SABADO TARDE

Name	Size	Modified
..		
.Rhistory	27 B	May 1, 2026, 1:05 AM
DATA		
LUCERO SISTEMATIZACION.Rproj	218 B	Jun 28, 2026, 3:06 AM
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pacientes.csv	1.1 KB	Jun 27, 2026, 2:40 PM
QUARTO		
SCRIPT		

Chunk 3

Console Terminal Background Jobs

R 4.5.3 C:/Users/LENOVO/OneDrive/Escritorio/LUCERO SISTEMATIZACION - SABADO TARDE/

```
> head(datos)
> summary(datos)
```

Paciente Edad IMC Fuerza_Muscular Equilibrio Medicamentos

Length:20 Min. :66.00 Min. :24.00 Min. :58.0 Min. :63.00 Min. :1.00

Class :character 1st Qu.:70.75 1st Qu.:26.75 1st Qu.:64.5 1st Qu.:69.75 1st Qu.:2.75

Mode :character Median :75.50 Median :29.00 Median :71.0 Median :75.50 Median :4.00

Mean :75.50 Mean :28.90 Mean :71.5 Mean :76.50 Mean :3.70

3rd Qu.:80.25 3rd Qu.:31.25 3rd Qu.:78.5 3rd Qu.:83.50 3rd Qu.:5.00

Max. :85.00 Max. :34.00 Max. :85.0 Max. :90.00 Max. :6.00

Vision Velocidad_Marcha

Min. :66.00 Min. :0.780

1st Qu.:71.75 1st Qu.:0.895

Median :78.00 Median :1.010

Mean :78.15 Mean :1.030

3rd Qu.:83.50 3rd Qu.:1.163

Max. :91.00 Max. :1.300

PASO 2

LUCERO SISTEMATIZACION - SABADO TARDE - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Render on Save ABC Render

Source Visual B T Normal Format Insert Table Outline

Package markdown required but is not installed. Install Don't Show Again

```
{r}
variables <- datos[,2:8]
head(variables)
```

Quarto Running Code

Environment History Connections Tutorial

R Global Environment

Data

datos	20 obs. of 8 variables
pca	List of 5
variables	20 obs. of 7 variables

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

C > Users > LENOVO > OneDrive > Escritorio > LUCERO SISTEMATIZACION - SABADO TARDE

Name	Size
..	
.Rhistory	27 B
DATA	
LUCERO SISTEMATIZACION.Rproj	218 B
pacientes_caidas.csv	614 B
pacientes.csv	1.1 KB
QUARTO	
SCRIPT	

Chunk 4

Console Terminal Background Jobs

R 4.5.3 C:/Users/LENOVO/OneDrive/Escritorio/LUCERO SISTEMATIZACION - SABADO TARDE/

```
> summary(variables)
```

Paciente Edad IMC Fuerza_Muscular Equilibrio Medicamentos

Length:20 Min. :66.00 Min. :24.00 Min. :58.0 Min. :63.00 Min. :1.00

Class :character 1st Qu.:70.75 1st Qu.:26.75 1st Qu.:64.5 1st Qu.:69.75 1st Qu.:2.75

Mode :character Median :75.50 Median :29.00 Median :71.0 Median :75.50 Median :4.00

Mean :75.50 Mean :28.90 Mean :71.5 Mean :76.50 Mean :3.70

3rd Qu.:80.25 3rd Qu.:31.25 3rd Qu.:78.5 3rd Qu.:83.50 3rd Qu.:5.00

Max. :85.00 Max. :34.00 Max. :85.0 Max. :90.00 Max. :6.00

Vision Velocidad_Marcha

Min. :66.00 Min. :0.780

1st Qu.:71.75 1st Qu.:0.895

Median :78.00 Median :1.010

Mean :78.15 Mean :1.030

3rd Qu.:83.50 3rd Qu.:1.163

Max. :91.00 Max. :1.300

```
> variables <- datos[,2:8]
> head(variables)
```

	Edad	IMC	Fuerza_M...	Equilibrio	Medicam...	Vision
	<int>	<int>	<int>	<int>	<int>	<int>
1	66	24	85	90	1	90
2	68	25	82	88	2	88
3	70	26	80	85	2	85
4	72	27	78	82	3	82
5	74	28	75	80	3	80
6	76	29	70	75	4	78

6 rows | 1-7 of 7 columns

PASO 3

Lucero Sistematización - Sábado Tarde - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Addins

Render on Save

Render

Run

Publish

Source Visual B I <> Normal <> Format <> Insert <> Table <> Outline

Package markdown required but is not installed. Install Don't Show Again

5	74	28	75	80	3	80
6	76	29	70	75	4	78

6 rows | 1-7 of 7 columns

{r}

variables.std <- scale(variables)

Chunk 5

Quarto

Quarto Running Code

Environment History Connections Tutorial

Import Dataset 134 MiB

Global Environment

Data

datos	20 obs. of 8 variables
pca	List of 5
variables	20 obs. of 7 variables
variables.std	num [1:20, 1:7] -1.606 -1.268 -0.93 -0.592 -0...

Files Plots Packages Help Viewer Presentation

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C > Users > LENOVO > OneDrive > Escritorio > LUCERO SISTEMATIZACION - SABADO TARDE

Name	Size	Modified
..		
.Rhistory	27 B	May 1, 2026, 1:05 AM
DATA		
LUCERO SISTEMATIZACION.Rproj	218 B	Jun 28, 2026, 3:06 AM
pacientes_caidas.csv	614 B	Jun 27, 2026, 3:02 PM
pacientes.csv	1.1 KB	Jun 27, 2026, 2:40 PM
QUARTO		
SCRIPT		

Console Terminal Background Jobs

R 4.5.3 - C:/Users/LENOVO/OneDrive/Escritorio/LUCERO SISTEMATIZACION - SABADO TARDE/

class : character

Mode : character

1st Qu.:70.75 1st Qu.:20.75 1st Qu.:04.5 1st Qu.:09.75 1st Qu.:2.75

Median :75.50 Median :29.00 Median :71.0 Median :75.50 Median :4.00

Mean :75.50 Mean :28.90 Mean :71.5 Mean :76.50 Mean :3.70

3rd Qu.:80.25 3rd Qu.:31.25 3rd Qu.:78.5 3rd Qu.:83.50 3rd Qu.:5.00

Max. :85.00 Max. :34.00 Max. :85.0 Max. :90.00 Max. :6.00

Vision Velocidad_Marcha

Min. :66.00 Min. :0.780

1st Qu.:71.75 1st Qu.:0.895

Median :78.00 Median :1.010

Mean :78.15 Mean :1.030

3rd Qu.:83.50 3rd Qu.:1.163

Max. :91.00 Max. :1.300

> variables <- datos[,2:8]

> head(variables)

> variables.std <- scale(variables)

PASO 4

Lucero Sistematización - Sábado Tarde - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Addins

Render on Save

Render

Run

Publish

Source Visual B I <> Normal <> Format <> Insert <> Table <> Outline

Package markdown required but is not installed. Install Don't Show Again

{r}

variables.std <- scale(variables)

{r}

pca <- prcomp(variables.std)

summary(pca)

Chunk 6

Quarto

Quarto Running Code

Environment History Connections Tutorial

Import Dataset 134 MiB

Global Environment

Data

datos	20 obs. of 8 variables
pca	List of 5
variables	20 obs. of 7 variables
variables.std	num [1:20, 1:7] -1.606 -1.268 -0.93 -0.592 -0...

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

C > Users > LENOVO > OneDrive > Escritorio > LUCERO SISTEMATIZACION - SABADO TARDE

Name	Size	Modified
..		
.Rhistory	27 B	May 1, 2026, 1:05 AM
DATA		
LUCERO SISTEMATIZACION.Rproj	218 B	Jun 28, 2026, 3:06 AM
pacientes_caidas.csv	614 B	Jun 27, 2026, 3:02 PM
pacientes.csv	1.1 KB	Jun 27, 2026, 2:40 PM
QUARTO		
SCRIPT		

Console Terminal Background Jobs

R 4.5.3 - C:/Users/LENOVO/OneDrive/Escritorio/LUCERO SISTEMATIZACION - SABADO TARDE/

1st Qu.:71.75 1st Qu.:0.895

Median :78.00 Median :1.010

Mean :78.15 Mean :1.030

3rd Qu.:83.50 3rd Qu.:1.163

Max. :91.00 Max. :1.300

> variables <- datos[,2:8]

> head(variables)

> variables.std <- scale(variables)

> pca <- prcomp(variables.std)

> summary(pca)

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.638	0.15397	0.08950	0.06330	0.05587	0.04644	0.02754
Proportion of Variance	0.994	0.00339	0.00114	0.00057	0.00045	0.00031	0.00011
Cumulative Proportion	0.994	0.99742	0.99857	0.99914	0.99958	0.99989	1.00000

PASO 5

Lucero Sistematización - Sábado Tarde - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Render on Save Render

Source Visual Normal Format Insert Table Outline

Package rmarkdown required but is not installed. Install Don't Show Again

variables

summary(pca)\$importance

	PC1	PC2	PC3	PC4	PC5
PC6	2.637848	0.1539671	0.08949886	0.06330438	0.05587189
PC7	0.04644196	0.02754011			
Standard deviation	0.994030	0.0033900	0.00114000	0.00057000	0.00045000
Proportion of Variance	0.994030	0.0033900	0.00114000	0.00057000	0.00045000
Cumulative Proportion	0.994030	0.9974200	0.99857000	0.99914000	0.99958000

Console

```
R - R 4.5.3 - C:/Users/LENOVO/OneDrive/Escritorio/LUCERO SISTEMATIZACION - SABADO TARDE/
> variables <- datos[,2:8]
> head(variables)
> variables.std <- scale(variables)
> pca <- prcomp(variables.std)
> summary(pca)
Importance of components:
              PC1      PC2      PC3      PC4      PC5      PC6      PC7
Standard deviation  2.638 0.15397 0.08950 0.06330 0.05587 0.04644 0.02754
Proportion of Variance 0.994 0.00339 0.00114 0.00057 0.00045 0.00031 0.00011
Cumulative Proportion 0.994 0.99742 0.99857 0.99914 0.99958 0.99989 1.00000
> summary(pca)$importance
              PC1      PC2      PC3      PC4      PC5      PC6      PC7
Standard deviation  2.637848 0.1539671 0.08949886 0.06330438 0.05587189 0.04644196 0.02754011
Proportion of Variance 0.994030 0.0033900 0.00114000 0.00057000 0.00045000 0.00031000 0.00011000
Cumulative Proportion 0.994030 0.9974200 0.99857000 0.99914000 0.99958000 0.99989000 1.00000000
>
```

Environment

History

Connections

Tutorial

R - Global Environment

Data

datos	20 obs. of 8 variables
pca	List of 5
variables	20 obs. of 7 variables
variables.std	num [1:20, 1:7] -1.606 -1.268 -0.93 -0.592 -0

Files

Plots

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Help

Viewer

Presentation

New Folder

New File

Delete

Rename

More

C:/Users/LENOVO/OneDrive/Escritorio/LUCERO SISTEMATIZACION - SABADO TARDE

..

.Rhistory

27 B

May 1, 2026, 1:05 AM

DATA

LUCERO SISTEMATIZACION.Rproj

218 B

Jun 28, 2026, 3:06 AM

pacientes_caidas.csv

614 B

Jun 27, 2026, 3:02 PM

pacientes.csv

1.1 KB

Jun 27, 2026, 2:40 PM

QUARTO

SCRIPT

PASO 6

Lucero Sistematización - Sábado Tarde - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Render on Save Render

Source Visual Normal Format Insert Table Outline

Package rmarkdown required but is not installed. Install Don't Show Again

variables

pca

datos

```
pca,
cex = 0.8,
col = c("red", "blue")
)
```

Console

```
R - R 4.5.3 - C:/Users/LENOVO/OneDrive/Escritorio/LUCERO SISTEMATIZACION - SABADO TARDE/
Standard deviation  2.638 0.15397 0.08950 0.06330 0.05587 0.04644 0.02754
Proportion of Variance 0.994 0.00339 0.00114 0.00057 0.00045 0.00031 0.00011
Cumulative Proportion 0.994 0.99742 0.99857 0.99914 0.99958 0.99989 1.00000
> summary(pca)$importance
              PC1      PC2      PC3      PC4      PC5      PC6      PC7
Standard deviation  2.637848 0.1539671 0.08949886 0.06330438 0.05587189 0.04644196 0.02754011
Proportion of Variance 0.994030 0.0033900 0.00114000 0.00057000 0.00045000 0.00031000 0.00011000
Cumulative Proportion 0.994030 0.9974200 0.99857000 0.99914000 0.99958000 0.99989000 1.00000000
>
```

Environment

History

Connections

Tutorial

R - Global Environment

Data

datos	20 obs. of 8
pca	List of 5
variables	20 obs. of 7
variables.std	num [1:20, 1:

Files

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Packages

Help

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Pre

New Folder

New File

Delete

C:/Users/LENOVO/OneDrive/Escritorio

..

.Rhistory

DATA

LUCERO SISTEMATIZACION.Rproj

pacientes_caidas.csv

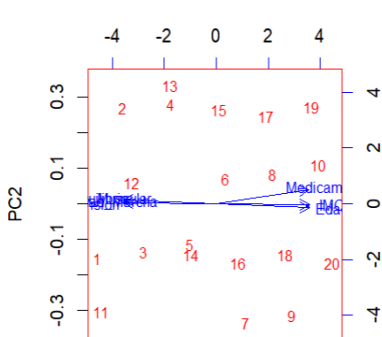
pacientes.csv

QUARTO

SCRIPT

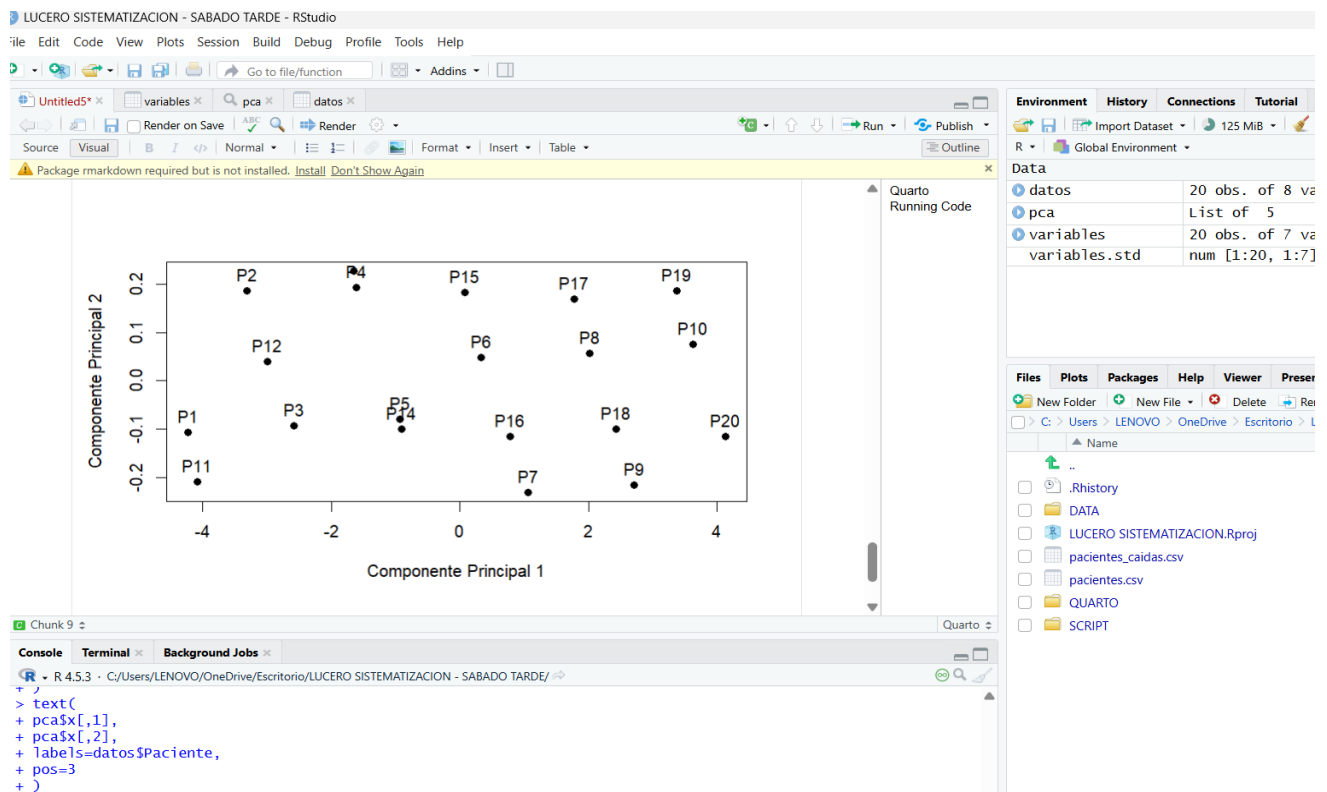
Quarto

Running Code



Chunk 8

UNICAMENTE PACIENTES



UNICAMENTE VARIABLES

